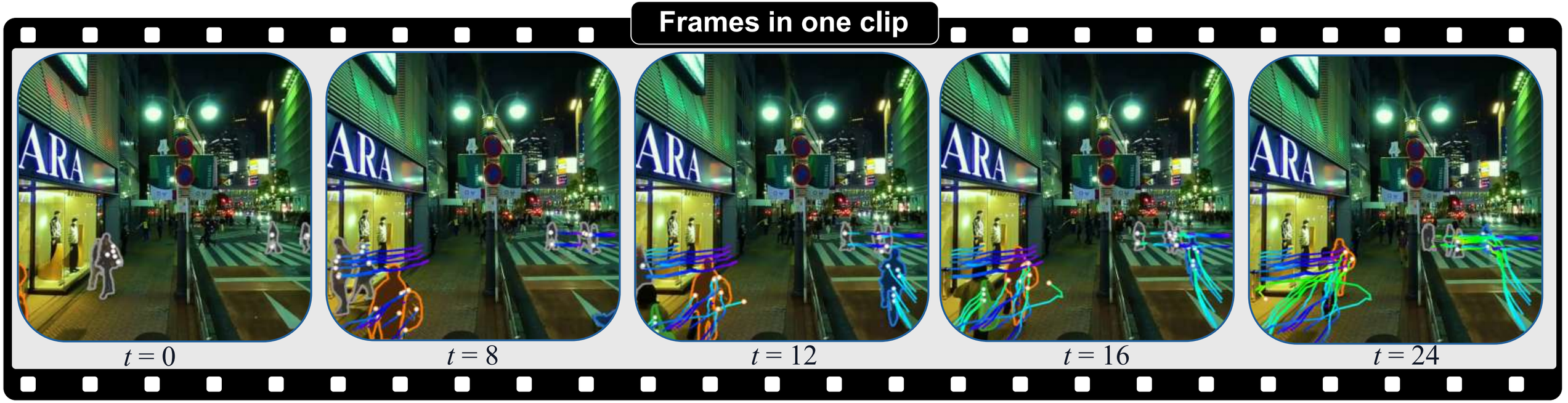


Question

In the world frame (ignoring camera motion), what was the average speed (in meters per second) of the bicycle between frame 12 and frame 16 of 34?

A. 1.7 **B. 4.3** **C. 3.2** **D. 2.3**

reference: 4.13 m/s

Trajectory tokens (serialized input to MLLM)

[scene] [the person] [part 1] [part 2] <traj> person <traj>

[the shopping bag] [the bicycle] [the camera]

<traj> bicycle <traj> <traj> camera <traj>

MLLM

Qwen3-VL-8B (frames only)

moved 1.0 m in 0.80 s

→ 1.29 m/s → A ✗

Frames + trajectory tokens

(2.7, 2.2, 6.4) m → (3.8, 2.8, 9.3) m

3.2 m in 0.80 s → 3.95 m/s → B ✓